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COURSE : Data handling and visualization

COURSE CODE:DSA0602

1.Hospital Patient Admission Analysis

CODE:

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
import squarify
np.random.seed(123)
departments = [
    "Cardiology",
    "Neurology",
    "Pediatrics",
    "Orthopedics",
    "Emergency"
]

admissions = pd.DataFrame({
    "Department": np.repeat(departments, 50),
    "Patients": np.random.randint(20, 150, 250)
})

print(admissions.head())
```

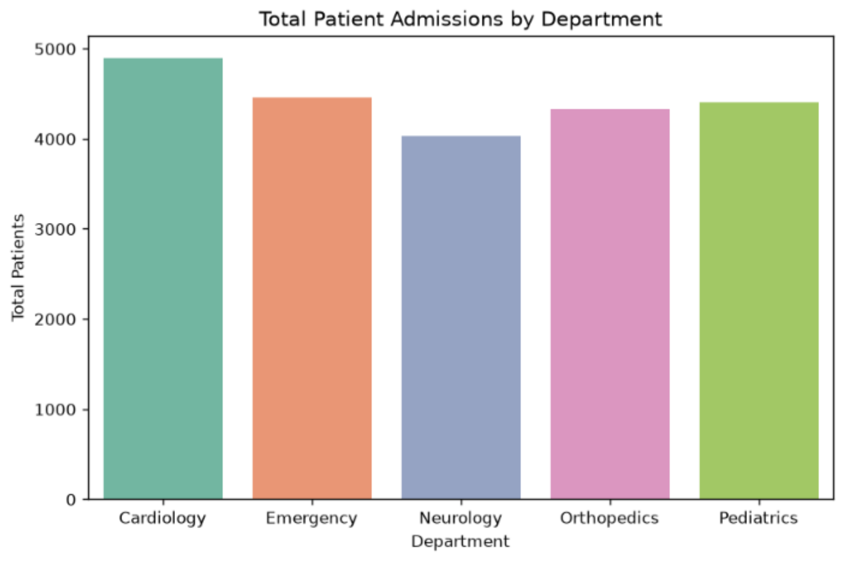
```python
deptTotal = admissions.groupby(
    "Department",
    as_index=False
)["Patients"].sum()

plt.figure(figsize=(8,5))

sns.barplot(
    data=deptTotal,
    x="Department",
    y="Patients",
    hue="Department",
    palette="Set2",
    legend=False
)

plt.title("Total Patient Admissions by Department")
plt.xlabel("Department")
plt.ylabel("Total Patients")

plt.show()

admissions["Month"] = np.tile(
    np.random.choice(
        ["Jan","Feb","Mar","Apr","May"],
        50
    ),
    5
```

```
)

pivot = admissions.pivot_table(
    values="Patients",
    index="Department",
    columns="Month",
    aggfunc="sum"
)

pivot.plot(
    kind="bar",
    stacked=True,
    figsize=(10,6)
)

plt.title("Department Admissions by Month")
plt.xlabel("Department")
plt.ylabel("Patients")

plt.show()

plt.figure(figsize=(10,6))

squarify.plot(
    sizes=deptTotal["Patients"],
    label=deptTotal["Department"],
    alpha=0.8
)
```

```python
plt.axis("off")

plt.title("Treemap of Department Workload")

plt.show()

plt.figure(figsize=(8,5))

sns.boxplot(
    data=admissions,
    x="Department",
    y="Patients",
    hue="Department",
    palette="Pastel1",
    legend=False
)

plt.title("Admission Variability by Department")

plt.show()

plt.figure(figsize=(8,5))

sns.histplot(
    admissions["Patients"],
    bins=20,
    kde=True,
```

```
    color="steelblue"
)

plt.title("Distribution of Patient Admissions")
plt.xlabel("Number of Patients")

plt.show()

print("Best Visualization: Bar Chart")
print("Reason: It clearly compares department workload.")
print("Box Plot is useful for identifying variability and outliers.")
print("Histogram shows overall patient distribution.")
print("Treemap displays department proportions.")
print("Stacked Bar Chart compares monthly admissions.")
```

OUTPUT:



2. E-Commerce Customer Spending Behaviour

CODE:

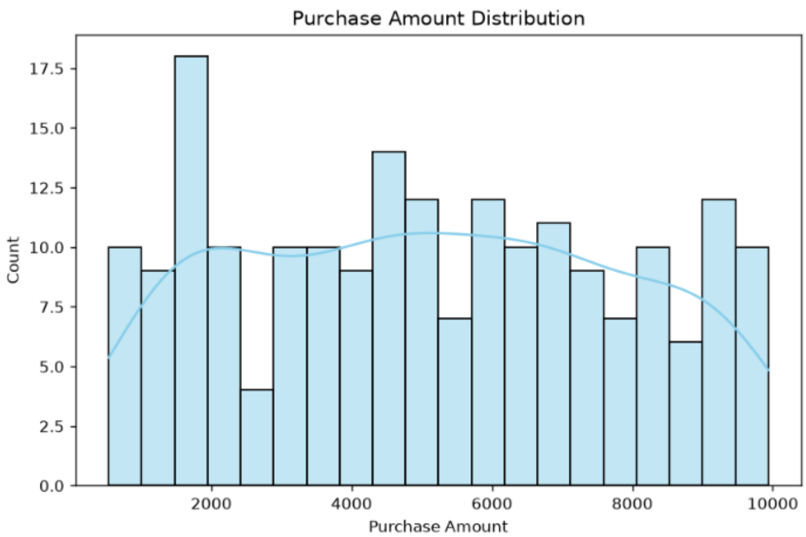
```
import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

import seaborn as sns

np.random.seed(123)

customers = pd.DataFrame({

    "PurchaseAmount": np.random.randint(500, 10000, 200),

    "OrderFrequency": np.random.randint(1, 25, 200),

    "Category": np.random.choice(

        ["Electronics", "Clothing", "Home", "Books", "Sports"],

        200

    )

})

print(customers.head())

plt.figure(figsize=(8,5))

sns.histplot(

    customers["PurchaseAmount"],

    bins=20,

    kde=True,

    color="skyblue"

)


plt.title("Purchase Amount Distribution")

plt.xlabel("Purchase Amount")

plt.ylabel("Count")
```

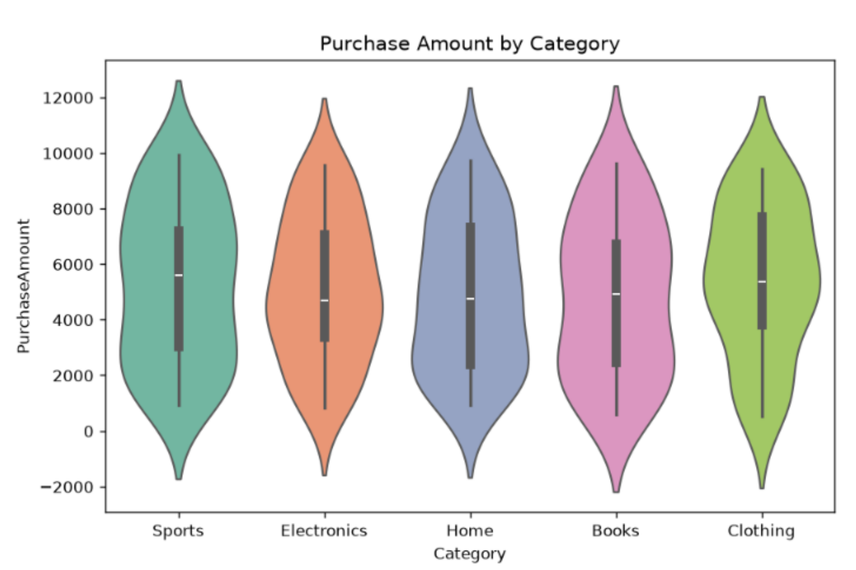
```
plt.show()
plt.figure(figsize=(8,5))
sns.violinplot(
    data=customers,
    x="Category",
    y="PurchaseAmount",
    hue="Category",
    palette="Set2",
    legend=False
)
plt.title("Purchase Amount by Category")
plt.show()
plt.figure(figsize=(8,5))
sns.boxplot(
    data=customers,
    x="Category",
    y="PurchaseAmount",
    hue="Category",
    palette="Pastel1",
    legend=False
)

plt.title("Purchase Amount by Category")

plt.show()

plt.figure(figsize=(8,5))
```

```
sns.kdeplot(
    customers["PurchaseAmount"],
    fill=True,
    color="green"
)

plt.title("Density Plot of Purchase Amount")
plt.xlabel("Purchase Amount")

plt.show()

categorySales = customers.groupby(
    "Category",
    as_index=False
)["PurchaseAmount"].mean()

plt.figure(figsize=(8,5))

sns.barplot(
    data=categorySales,
    x="Category",
    y="PurchaseAmount",
    hue="Category",
    palette="viridis",
    legend=False
)

plt.title("Average Purchase Amount by Category")
```

```
plt.xlabel("Category")
plt.ylabel("Average Purchase")

plt.show()

print("Best Visualization: Box Plot")
print("Reason: It clearly identifies high-value customers, spending variability and outliers.")
print("Histogram shows purchase distribution.")
print("Violin plot shows data distribution shape.")
print("Density plot shows spending trends.")
print("Bar chart compares average spending across categories.")
```

OUTPUT:





3.University Examination Performance Dashboard

CODE:

```
import numpy as np

import pandas as pd

import seaborn as sns

import matplotlib.pyplot as plt

import plotly.express as px


np.random.seed(123)


students = pd.DataFrame({

    "Department": np.random.choice(

        ["CSE", "ECE", "EEE", "Mechanical", "Civil"],

        200

    ),

    "Marks": np.random.randint(35, 101, 200),

    "Class": np.random.choice(

        ["A", "B", "C", "D"],
```

```
    200
  )
})

print(students.head())

plt.figure(figsize=(8,5))

sns.boxplot(
  data=students,
  x="Department",
  y="Marks",
  hue="Department",
  palette="Set2",
  legend=False
)

plt.title("Marks Distribution by Department")

plt.show()

plt.figure(figsize=(8,5))

sns.histplot(
  students["Marks"],
  bins=15,
  kde=True,
  color="steelblue"
```

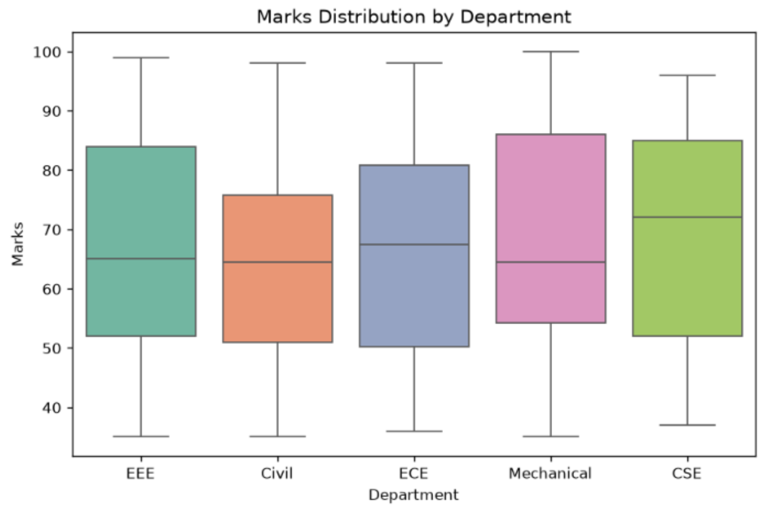
```
)

plt.title("Marks Distribution")

plt.xlabel("Marks")

plt.show()

classAverage = students.groupby(
    "Class",
    as_index=False
)["Marks"].mean()

plt.figure(figsize=(7,5))

sns.barplot(
    data=classAverage,
    x="Class",
    y="Marks",
    hue="Class",
    palette="viridis",
    legend=False
)

plt.title("Average Marks by Class")

plt.show()
```

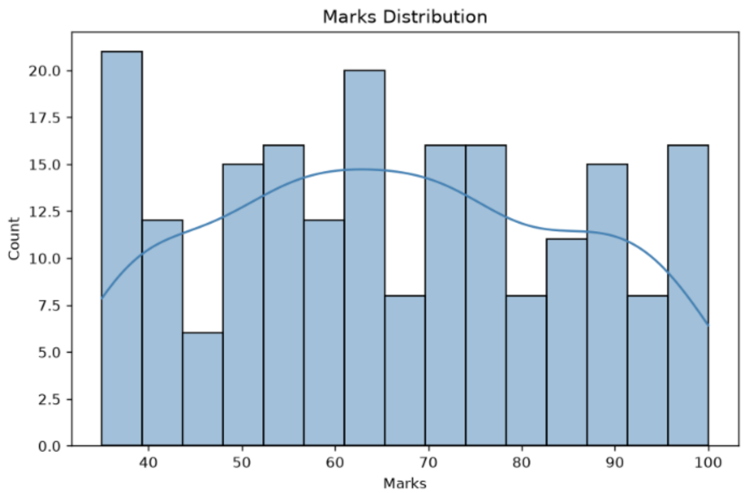
```python
departmentAverage = students.groupby(
    "Department",
    as_index=False
)["Marks"].mean()

fig = px.bar(
    departmentAverage,
    x="Department",
    y="Marks",
    color="Department",
    title="Average Marks by Department"
)

fig.show()

fig = px.box(
    students,
    x="Department",
    y="Marks",
    color="Department",
    title="Department Wise Marks Distribution"
)

fig.show()
print("Evaluation")
print("1. Box Plot identifies score distribution and outliers.")
print("2. Histogram shows overall mark distribution.")
print("3. Bar Chart compares average marks among classes.")
```

print("4. Plotly charts provide interactive analysis.")

print("Recommendation: Use the Box Plot and Plotly Bar Chart in the academic dashboard because they clearly show performance differences and outliers.")

OUTPUT:





4. Climate and Rainfall Distribution Study

CODE:

import numpy as np

import pandas as pd

import matplotlib.pyplot as plt

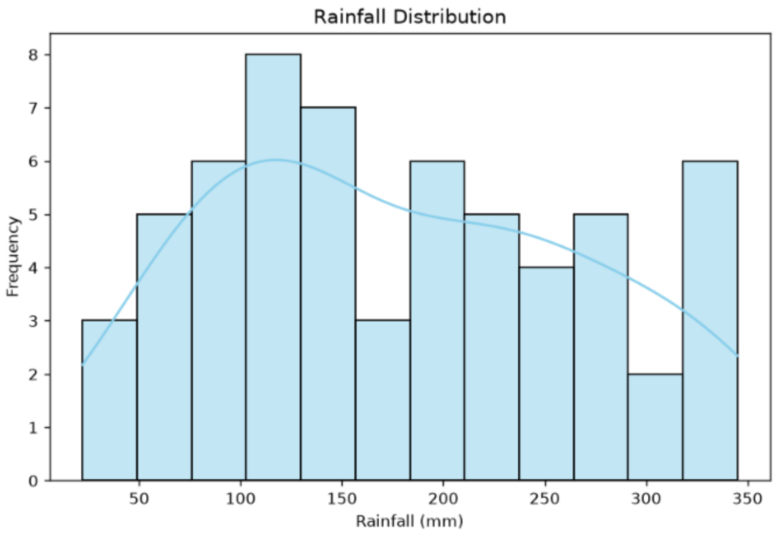
import seaborn as sns

```
np.random.seed(123)

climate = pd.DataFrame({
    "City": np.repeat(
        ["Chennai", "Bangalore", "Hyderabad", "Mumbai", "Delhi"],
        12
    ),
    "Month": list(range(1, 13)) * 5,
    "Rainfall": np.random.randint(20, 350, 60),
    "Temperature": np.random.randint(18, 42, 60),
    "Humidity": np.random.randint(40, 95, 60)
})
print(climate.head())
plt.figure(figsize=(8,5))
sns.histplot(
    climate["Rainfall"],
    bins=12,
    kde=True,
    color="skyblue"
)

plt.title("Rainfall Distribution")
plt.xlabel("Rainfall (mm)")
plt.ylabel("Frequency")
plt.show()
plt.figure(figsize=(8,5))

sns.kdeplot(
```

```
    climate["Rainfall"],
    fill=True,
    color="green"
)

plt.title("Density Plot of Rainfall")
plt.xlabel("Rainfall (mm)")
plt.show()
plt.figure(figsize=(8,5))
sns.boxplot(
    data=climate,
    x="City",
    y="Rainfall",
    hue="City",
    palette="Set2",
    legend=False
)

plt.title("Rainfall Distribution by City")

plt.show()

averageRainfall = climate.groupby(
    "City",
    as_index=False
)["Rainfall"].mean()

plt.figure(figsize=(8,5))
```

```
sns.barplot(
    data=averageRainfall,
    x="City",
    y="Rainfall",
    hue="City",
    palette="viridis",
    legend=False
)
plt.title("Average Rainfall by City")
plt.xlabel("City")
plt.ylabel("Average Rainfall (mm)")

plt.show()

print("Evaluation")
print("Histogram: Shows rainfall frequency distribution.")
print("Density Plot: Displays the rainfall pattern smoothly.")
print("Box Plot: Best for comparing rainfall variation and detecting outliers.")
print("Bar Chart: Best for comparing average rainfall among cities.")

print("Recommendation")
print("Box Plot is the most suitable visualization for environmental researchers because it clearly shows seasonal variation, spread and outliers.")
```

OUTPUT:



5. Bank Loan Risk Assessment

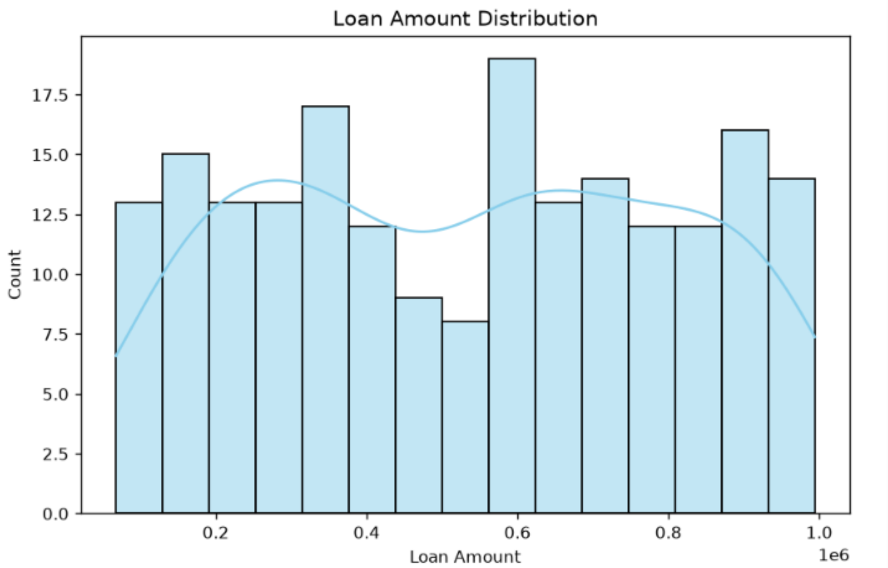
CODE:

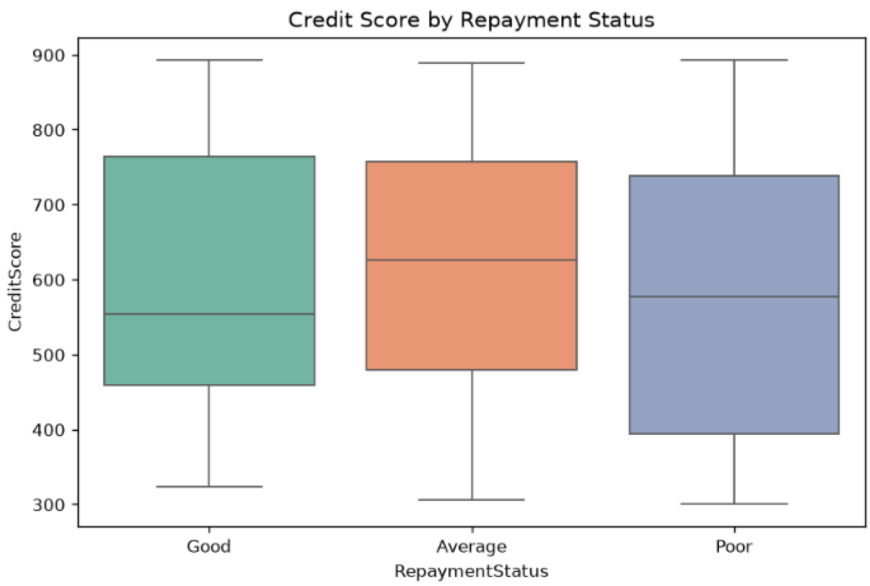
```
import numpy as np
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
import plotly.express as px

np.random.seed(123)

loan = pd.DataFrame({
    "LoanAmount": np.random.randint(50000, 1000000, 200),
    "Income": np.random.randint(20000, 150000, 200),
    "CreditScore": np.random.randint(300, 900, 200),
    "RepaymentStatus": np.random.choice(
        ["Good", "Average", "Poor"],
        200
    )
})
```

```
print(loan.head())

plt.figure(figsize=(8,5))

sns.histplot(
    loan["LoanAmount"],
    bins=15,
    kde=True,
    color="skyblue"
)

plt.title("Loan Amount Distribution")
plt.xlabel("Loan Amount")
plt.ylabel("Count")
plt.show()
plt.figure(figsize=(8,5))sns.boxplot(
    data=loan,
    x="RepaymentStatus",
    y="CreditScore",
    hue="RepaymentStatus",
    palette="Set2",
    legend=False
)

plt.title("Credit Score by Repayment Status")
plt.show()
fig = px.scatter(
```

```
    loan,
    x="Income",
    y="LoanAmount",
    color="RepaymentStatus",
    size="CreditScore",
    title="Customer Loan Risk Assessment"
)

fig.show()
fig = px.histogram(
    loan,
    x="CreditScore",
    color="RepaymentStatus",
    nbins=20,
    title="Credit Score Distribution"
)

fig.show()

print("Evaluation")
print("Histogram shows the distribution of loan amounts.")
print("Box Plot identifies outliers and repayment risk.")
print("Scatter Plot reveals the relationship between income and loan amount.")
print("Plotly Histogram compares credit scores across repayment groups.")

print("Recommendation")
print("Scatter Plot is the best visualization for the management dashboard because it clearly highlights customer segments, repayment risk, and potential outliers.")
```

OUTPUT:


Loan Amount Distribution
17.5
15.0
12.5
10.0
7.5
5.0
2.5
0.0
Count
0.2
0.4
0.6
0.8
1.0
1e6
Loan Amount


Credit Score by Repayment Status
900
800
700
600
500
400
300
CreditScore
Good
Average
Poor
RepaymentStatus